

EXECUTIVE BUSINESS PROPOSAL

PatientTriage.ai

Continuous Emergency Department Safety Monitoring & Hybrid Decision Support

See risk before it becomes a crisis — shifting triage from an isolated intake snapshot into active waiting-room surveillance.



Continuous Radar™

Real-time longitudinal vital tracking & wait-time threshold surveillance.



Hybrid Safety Fusion

Deterministic safety floors enforce unbreachable red flags over advisory ML.



100% Clinician Authority

Human-in-the-loop governance with mandatory audit reason logging.

PROTOTYPE REGULATORY NOTICE: Clinical decision-support prototype evaluated on synthetic demonstration benchmarks. Recommendations are strictly advisory. Licensed clinicians retain 100% decision authority. Formal CDSCO / CE / FDA SaMD review required prior to clinical deployment.

Executive Summary & Core Strategic Pillars

The Emergency Department Paradox

- Static Intake Snapshot: Conventional triage assigns an acuity score once at minute zero ($t=0$).
- Dynamic Physiology: Patients wait 30 minutes to 4+ hours in crowded waiting rooms where conditions evolve unnoticed.
- The Safety Gap: Initially moderate cases (ESI-3) can silently deteriorate into life-threatening states before bedside exam.
- The Solution: PatientTriage.ai bridges this gap with continuous physiological surveillance and hybrid decision support.



1. Continuous Surveillance

Waiting-Room Radar™ tracks longitudinal vital deltas and wait times in real time.



2. Hybrid Safety Fusion

Deterministic rules establish non-downgradable CRITICAL safety floors over ML predictions.



3. Point-of-Care Explainability

Live SHAP feature contributions display exact mathematical drivers behind advisory outputs.



4. 100% Clinician Authority

Human-in-the-loop control with mandatory override reason tracking and audit trails.

Problem Framing: The Waiting-Room Safety Gap



Silent Decompensation

Occult sepsis, evolving stroke, and atypical coronary syndromes progress without overt distress during queue delays.



Cognitive Saturation

Triage staff managing 40+ waiting patients cannot manually track time limits or recalculate vital trendlines.



Unsafe Normal Imputation

Rushed staff frequently assume omitted vitals are normal, violating the clinical rule that UNKNOWN $\neq$ NORMAL.

The Core Clinical Reality: Initial Risk $\neq$ Permanent Risk

Patient Risk(t) = f (Vitals(t), Symptoms(t), Elapsed Wait Time (t - t₀), Δ Physiology(t))

- Conventional Triage assumes Δ Physiology = 0, creating a dangerous operational blind spot.
- PatientTriage.ai continuously recalculates risk trajectory, turning passive waiting rooms into actively monitored zones.

Solution Overview: 5-Component Clinical Architecture

1. Structured Intake

Rapid digital entry of vitals, chief complaints with negation protection, medical history, and clinical cues.

2. Data Quality

UNKNOWN $\neq$ NORMAL
governance: unrecorded vitals reduce completeness and elevate model uncertainty.

3. Safety Rules

Deterministic red flags ($\text{SpO}_2 < 88\%$, $\text{SBP} < 80$, Stridor) establish unbreachable CRITICAL safety floors.

4. Advisory ML

In-browser XGBoost evaluates 71 domain features (shock index, pulse pressure, interactions) + SHAP.

5. Radar™ Surveillance

Continuous monitoring detects vital deltas and wait-cap limits under 100% clinician authority.

Why the Hybrid Model: Rules + ML + Clinician Authority

Deterministic Rules Only

Safety Floor, Low Nuance

- ✓ 100% reliable red-flag enforcement
- ✓ Verifiable, deterministic safety floors
- ✗ Inflexible on atypical presentations
- ✗ 44.71% synthetic accuracy (misses nuance)
- ✗ Cannot recognize multi-variable interactions

Pure Machine Learning

High Sensitivity, Fragile

- ✓ Recognizes complex clinical patterns
- ✓ 98.87% sensitivity on acute cases
- ✗ Vulnerable to missing/adversarial data
- ✗ Can hallucinate lower acuity on red flags
- ✗ Unsafe as autonomous decision authority

PatientTriage.ai Hybrid

The Optimal Clinical Balance

- ✓ Deterministic rules establish floor
- ✓ XGBoost advises on subtle interactions
- ✓ ML can UPGRADE for safety, never downgrade
- ✓ UNKNOWN ≠ NORMAL uncertainty handling
- ✓ 100% Licensed Clinician final authority

Target Users & 4-Stakeholder Value Ecosystem



Triage & Staff Nurses

Operational Relief at Triage

- Eliminates manual wait-time tracking across 40+ waiting patients
- Live data completeness gauge prevents unrecorded vital omissions
- Objective deterioration alerts validate nursing intuition during surges



Emergency Physicians

Instant Longitudinal Context

- Pre-arrival risk decomposition and local SHAP explainability
- Evidence Diff displays exact vital trajectory since intake
- 1-click override modal with structured clinical reason capture



ED Operations & Directors

Department-Wide Visibility

- Real-time queue acuity distribution and surge pressure monitoring
- 100% audit log coverage of priority overrides and safety floors
- Configurable facility profiles adapt wait caps to ED staffing



Hospital C-Suite & Risk

Liability & Capacity Protection

- Mitigates waiting-room sentinel events and malpractice exposure
- Reduces Left Without Being Seen (LWBS) rate via active queue care
- Improves ED bed turnover and reduces avoidable ICU escalations

Product Experience & Clinical Operations Workflow

Step 1: Patient Intake

Structured capture of vitals, complaints, history, and cues. UNKNOWN $\neq$ NORMAL standard penalizes omitted data.

Step 2: Dual Hybrid Scoring

Deterministic engine checks hard red flags. XGBoost evaluates 71 features. Safety floor gate blocks downgrades.

Step 3: Command Center

Live queue displays risk scores, model probability, uncertainty levels, and wait-time timers with fast filtering.

Step 4: Waiting-Room Radar™

Continuous surveillance tracks longitudinal vital deltas and triggers REASSESS / ESCALATE alerts on deterioration.

Step 5: Review & Override

Physician inspects SHAP feature attributions, reviews Evidence Diff, and confirms or overrides priority with logged reason.

Business Value Pillars & Pilot Hypotheses

18–25%

Time-to-Intervention

Accelerates critical bedside escalation for occult sepsis and silent hypoxemia.

30–40%

Liability Risk Reduction

Eliminates unmonitored waiting-room sentinel events with 100% audit logging.

12–15%

ED Bed Turnover Uplift

Reduces queue bottlenecks and Left Without Being Seen (LWBS) rates.

20–30%

Cognitive Load Reduction

Automates longitudinal trend calculation and wait-cap surveillance.

Pilot Value Hypotheses (Targets for Clinical Pilot Studies)

- Deterioration Detection Rate: Radar surveillance catches 100% of vital drops (SpO₂ drop $\geq 3\%$, HR spike ≥ 15) before queue threshold expiry.
- LWBS Rate Reduction: Active waiting-room engagement and transparent monitoring status reduce LWBS by 15–20% in high-volume urban EDs.
- Complete Audit Defense: 100% structured override capture protects clinical staff and hospital legal counsel during retrospective reviews.

Competitive Differentiation Matrix

How PatientTriage.ai Compares to Existing Clinical Systems

DIMENSION	LEGACY ESI / MTS	EHR TRIAGE MODULES	BLACK-BOX AI	PATIENTTRIAGE.AI
Triage Model Surveillance	Static Snapshot (t=0)	Static Charting (t=0)	Static Model Score	Continuous Radar™
Deterioration Alerts Deltas & Diff	Manual Periodic Check	Static Threshold Alert	Model Score Only	Longitudinal Vital
Safety Floor Floor (Invariant)	Subjective Human Floor	Basic Threshold Alert	None (ML Can Downgrade)	Deterministic Safety
Missing Data Rule Uncertainty Gate	Assumes Normal (Unsafe)	Omits Feature	Mean Imputation	UNKNOWN ≠ NORMAL
Explainability Decomposition	None	None	Black-Box Probabilities	Live Point-of-Care SHAP
Human Control Reason Log	Nurse Decides Alone	Nurse Decides Alone	Unclear Authority	100% Clinician Authority +

Responsible AI & Clinical Governance Guardrails

AI Assists. Safety Rules Protect. Clinicians Decide.

Clinical decision support with clear, verifiable boundaries — no autonomous prescriptions or unsupervised action.



100% Clinician Authority

Human-in-the-loop: Clinician retains absolute final decision authority. 1-click overrides with mandatory reason codes.



UNKNOWN ≠ NORMAL Standard

Omitted vitals or zero history increase model uncertainty penalties rather than assuming benign baselines.



Immutable Audit Trail

Every priority assignment, uncertainty escalation, model recommendation, and clinician override is permanently logged.



Zero PHI Cloud Egress

In-browser edge inference executes tree ensembles directly in TypeScript on local client device without cloud streaming.

Phased Deployment Strategy & Validation Gates

Phase 1: Retrospective Shadow

Months 1–3 • Offline Analysis

- Ingest historical de-identified ED cohorts (10,000+ records)
- Evaluate concordancy against attending physician notes
- Confirm zero false-negative red-flag downgrades
- Validation Gate: >95% Critical sensitivity on holdouts

Phase 2: Live Silent Observer

Months 4–6 • Parallel Pipeline

- Silent execution alongside triage nurse without UI display
- Track Waiting-Room Radar vital delta capture in real time
- Monitor alert burden and optimize threshold dampeners
- Validation Gate: Alert fatigue rate <5% per shift

Phase 3: Active Advisory Pilot

Months 7–12 • Live CDS Pilot

- Advisory deployment in a single pilot emergency department
- Clinicians retain 100% authority with mandatory override logging
- Measure time-to-reassess reduction and override rate (10–15%)
- Validation Gate: Institutional Review Board (IRB) signoff

Product Roadmap: From Prototype to Enterprise Triage

Phase 1: Prototype Foundation (COMPLETED)

Hybrid decision engine, 71 domain features, in-browser XGBoost inference, UNKNOWN $\neq$ NORMAL data completeness scoring, deterministic safety floors.

Phase 2: Continuous Radar & Alerting (COMPLETED)

Longitudinal vital tracking, Evidence Diff component, Why Now? alert cards, surge mode simulation, 11 preconfigured demo clinical scenarios.

Phase 3: EHR Interoperability (Q1–Q2)

Bi-directional HL7 FHIR v4.0 integration, Epic/Cerner SMART-on-FHIR launch, automated vitals polling from waiting-room kiosks and bedside monitors.

Phase 4: Multi-Center Enterprise Scaling (Q3–Q4)

Multi-hospital cluster management, federated model calibration without centralized PHI aggregation, CDSCO/FDA SaMD submission dossier.

Comprehensive Risk & Mitigation Matrix

1. Automation Bias (Over-Reliance)

Risk: Clinicians might passively accept AI suggestions without independent clinical assessment.

Mitigation: Mandatory reason capture on overrides; UI prominently states 'Advisory Support'; deterministic red-flag overrides enforce human signoff.

2. Alert Fatigue from Radar Notifications

Risk: Excessive reassessment alerts during crowded shifts could cause staff to ignore warnings.

Mitigation: Configurable threshold dampeners; delta magnitude thresholds (SpO₂ drop $\geq 3\%$, HR ≥ 15); priority sorting in Waiting-Room Radar.

3. Missing Data & Unrepresented Patients

Risk: Patients with zero medical history or unmeasured vitals could receive improper triage.

Mitigation: UNKNOWN $\neq$ NORMAL standard: unmeasured fields elevate uncertainty and block acuity downgrades.

4. Network Downtime & EHR Disruption

Risk: Hospital network failures could interrupt cloud-dependent decision tools.

Mitigation: In-browser edge inference engine runs 100% locally with zero external API dependencies during patient care.

Commercialization Model & Illustrative ROI Framework

Tier 1: Community Hospital ED

\$2,500 – \$4,000 / month

Up to 30,000 annual visits. Core decision support, Waiting-Room Radar, standard EHR connector, audit log.

Tier 2: Regional Trauma Center

\$5,000 – \$8,500 / month

30,000–80,000 annual visits. Bi-directional FHIR integration, custom facility profiles, surge management.

Tier 3: Academic Health Network

Enterprise Contract

Multi-hospital network, enterprise analytics, federated model calibration, dedicated CDS compliance manager.

Illustrative Hospital ROI Framework (50,000 Annual Visits)

- LWBS Revenue Recovery: Reducing LWBS rate by 1.2% recovers ~600 patients/year (\$180,000+ net revenue).
- Avoidable ICU Transfer Savings: Early detection of occult sepsis in waiting room prevents 12–18 unplanned ICU escalations (\$240,000+ cost avoidance).
- Malpractice Risk Reserve: Demonstrated audit trail compliance and zero waiting-room sentinel events lower liability reserves (\$75,000/year).
- Nurse Retention & Overtime: Automating surveillance saves 350+ nursing overtime hours/year (\$35,000).

**TOTAL ESTIMATED VALUE: \$530,000 / year (vs
~\$72,000 annual software fee → 7.3× First-Year ROI)**

Prototype Evidence: Synthetic Holdout Benchmarks

90.79%

Overall Accuracy

5-class acuity classification on 2,250 synthetic holdout test records (vs 44.71% rule baseline).

88.20%

Macro F1-Score

Balanced precision & recall across all five triage acuity tiers without class bias.

95.99%

Critical Recall

High sensitivity on simulated life-threatening presentations (anaphylaxis, shock).

98.87%

High+Crit Sensitivity

Queue safety coverage: XGBoost catches complex multi-variable interactions.

Technical Feasibility & Synthetic Validation Stance

- Fast Edge Inference: In-browser tree traversal in TypeScript evaluates 71 features in <10ms with zero Python server dependencies.
- SMART-on-FHIR Readiness: Standardized JSON schemas map directly to FHIR Observation, Condition, and Encounter resources.
- 3 Hospital Profiles: Trauma Center, Community Hospital, and Rural ED presets adapt wait thresholds to staffing levels.
- Mandatory Clinical Disclaimer: All benchmark metrics are computed on synthetic demonstration datasets. Formal retrospective and prospective clinical trials are required before clinical deployment.

STRATEGIC CONCLUSION

Transforming Emergency Triage into Continuous Safety

PatientTriage.ai bridges the waiting-room safety gap by pairing deterministic physiological safety rules with advisory machine learning under absolute clinician authority.

Safety First

Deterministic safety floors prevent algorithmic downgrade of red-flag presentations. UNKNOWN $\neq$ NORMAL ensures missing data is never assumed benign.

Dynamic Radar™

Continuous surveillance replaces static snapshots, tracking longitudinal vital trends and alert diffs to prevent waiting-room decompensation.

Clinician Authority

100% human-in-the-loop control with structured override tracking, local SHAP explainability, and immutable audit logging.

Next Strategic Steps: Multi-Center Retrospective Validation & Pilot Onboarding

Public Repository: https://github.com/sahilsheoran999/patient_triage • Live Functional Prototype Available